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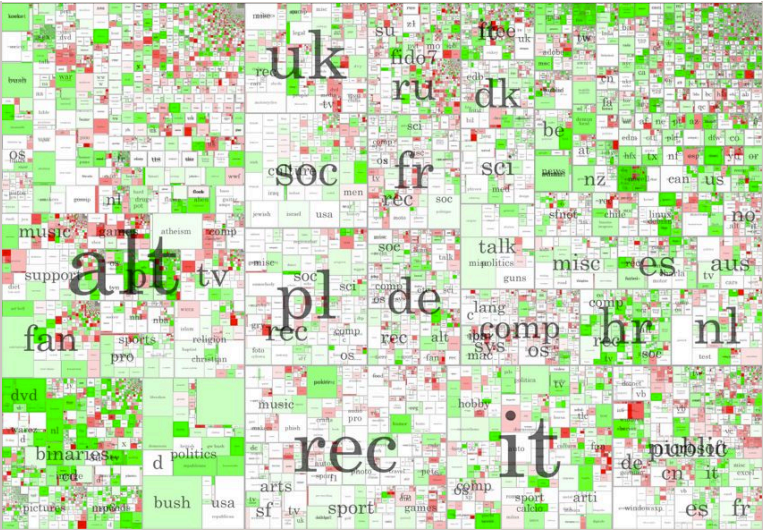
Science Maps Explore New Ways of Displaying Information [Slide Show]

A traveling, evolving exhibit of science maps will eventually contain 100 entries from around the world

By John Matson



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Smith, Marc and Danyel Fisher. 2004. Treemap View of 2004 Usenet Returnees, Redmond WA. Courtesy of Community Technologies Group, Microsoft Research. In Katy Börner & Deborah MacPherson (Eds.), 1st Iteration (2005): The Power of Maps, Places and Spaces: Mapping Science. <http://scimaps.org>

NET USE:

This 2005 map categorizes a massive amount of activity in Usenet newsgroups, a system of subject-specific Internet forums. Each rectangle represents one of nearly 200,000 newsgroups and is sized in proportion to the number of individuals who posted at least twice to that newsgroup in 2004. The hierarchical structure of Usenet is captured by overlapping labels—the parent "alt" label covers its subgroups such as "alt.tv" and "alt.politics," which in turn have their own subgroups nested within. Green rectangles represent newsgroups that grew in size from the prior year; red rectangles represent groups in decline.

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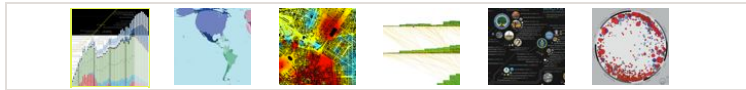


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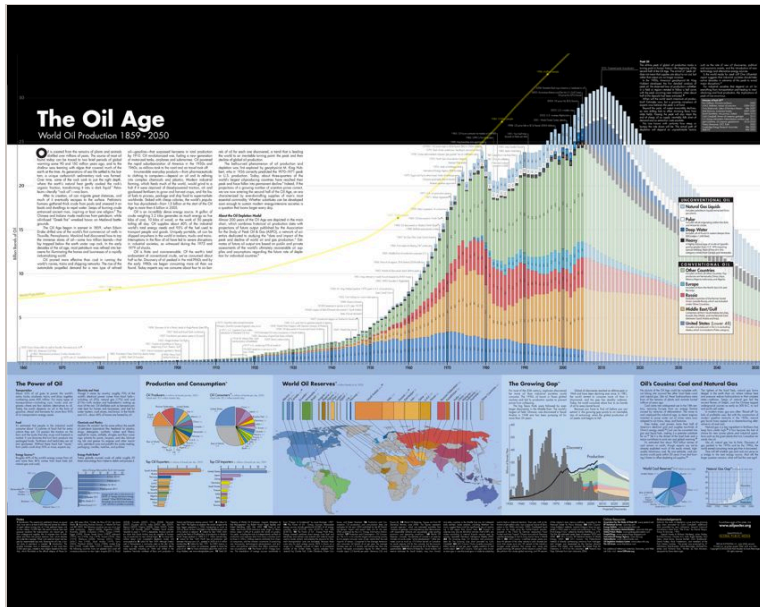
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Bracken, Rob (Writer), Dave Menninger (Graphic Artist), Michael Poremba (Statistician) and Richard Katz (Catalyst). 2006. [CLICK TO ENLARGE +](#)
The Oil Age: World Oil Production 1859-2050. San Francisco, CA. Courtesy of San Francisco Informatics. In Katy Börner & Julie M. Davis (Eds.), 3rd
Iteration (2007). The Power of Forecasts, Places and Spaces: Mapping Science. <http://scimaps.org>

A NEW AGE?

This map charts humanity's long and involved history with petroleum, tracing the world's swelling demand for oil alongside its swelling population. The production data begin in 1859 when, the map's authors report, the oil age began with the drilling of one of the world's first commercial oil wells in Pennsylvania. In 2006, when the map was produced, the authors forecast that 2010 would be the year of so-called peak oil—when the 150-year trend of increasing oil production comes to an end as discoveries of new reserves can no longer make up for declining production at existing wells. The arrival date of peak oil remains hotly debated.

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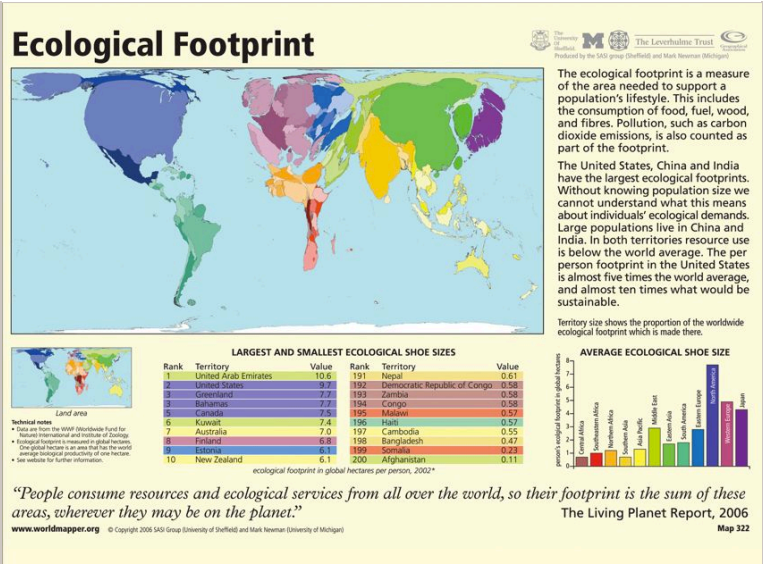
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Dorling, Danny, Mark Newman, Graham Allsopp, Anna Barford, Ben Wheeler, John Pritchard and David Dorling, 2006. Ecological Footprint. Sheffield, UK and Ann Arbor, MI. Courtesy of the University of Sheffield and University of Michigan. In Katy Börner & Elisha F. Hardy (Eds.), 4th Iteration (2008): Science Maps for Economic Decision Makers, *Places and Spaces: Mapping Science*. <http://scimaps.org>

IF THE SHOE FITS:

Each country on this manipulated world map has been resized according to its consumption of resources and its production of waste—its so-called ecological footprint. The U.S., which has the second-largest per capita footprint in the world, swells well beyond its true size, whereas the continent of Africa is shrunk almost beyond recognition. (The United Arab Emirates take the dubious honor of largest footprint per person.) Populous China and India have massive footprints as nations even though neither ranks in the top 10 of per capita consumption and waste. The resizing method, co-authored by Mark Newman of the University of Michigan at Ann Arbor, produces what is called a cartogram; for more, [see our slide show of Newman's electoral cartograms](#) from the 2008 U.S. presidential race.

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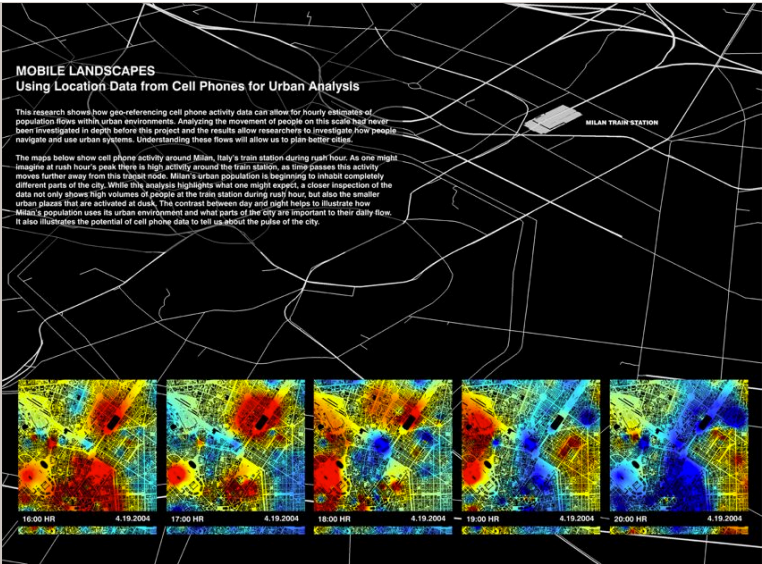
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Williams, Sarah, Carlo Ratti and Riccardo Maria Pulselli. 2006. Mobile Landscapes: Using Location Data from Cell Phones for Urban Analysis. Boston, MA. Courtesy of MIT SENSEable City Laboratory. In Katy Börner & Elisha F. Hardy (Eds.), 5th Iteration (2009): Science Maps for Science Policy Makers. *Places and Spaces: Mapping Science*. <http://scimaps.org>

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POCKET BEACONS:
Geolocation information from cell phones in Milan, Italy, provided a fine-grained estimate of human migration within the city on a single day in 2004. Over the course of four hours during the evening commute, the cell phone data, provided to the map's authors through a collaboration with Vodafone, a major European cell service provider, show how human density becomes concentrated at the train station and then disperses. The authors note that similar studies could aid urban planners in tracking the routes that city dwellers actually follow, and for identifying which public spaces find the most use.

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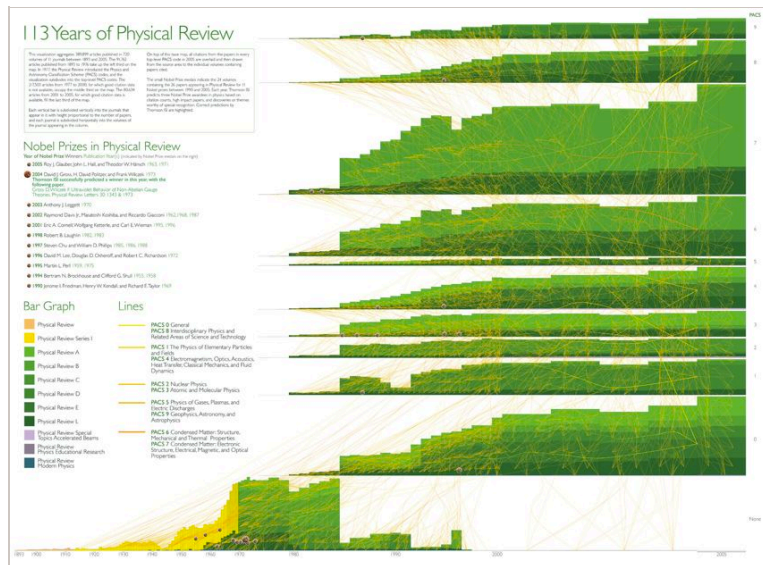
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Herr II, Bruce W., Russell Jackson Duhon (Data Mining & Visualization), Elisha F. Hardy (Graphic Design), Shasikant Penumarthi (Data Preparation) and Katy Börner (Concept), 2007. 113 Years of Physical Review, Bloomington, IN. Courtesy of Indiana University. In Katy Börner & Julie M. Davis (Eds.), 3rd iteration (2007): The Power of Forecasts, *Places and Spaces: Mapping Science*. <http://scimaps.org>

JOURNAL-ISM:

Nearly 400,000 scholarly papers, representing more than a century of work in *Physical Review* journals, are plotted and linked on this 2007 map. The papers are categorized by broad taxonomic groups based on subject matter—each subject area makes up one of the 11 rows of data. (The bottom row, which represents uncategorized papers, extends further back in time because the classification system, known as PACS, was not introduced until 1977.) The horizontal axis separates the publication years, between 1893 and 2005, whereas the vertical height of each bar represents how many papers were published that year in that particular subject area. The third bar from the top, representing condensed-matter physics, packs the most papers. Each bar is color coded to separate contributions from the 11 individual journals. The web of yellow strands crisscrossing the chart connects published articles to the reference works cited therein, and medallions indicate papers that would later earn Nobel Prizes for their authors.

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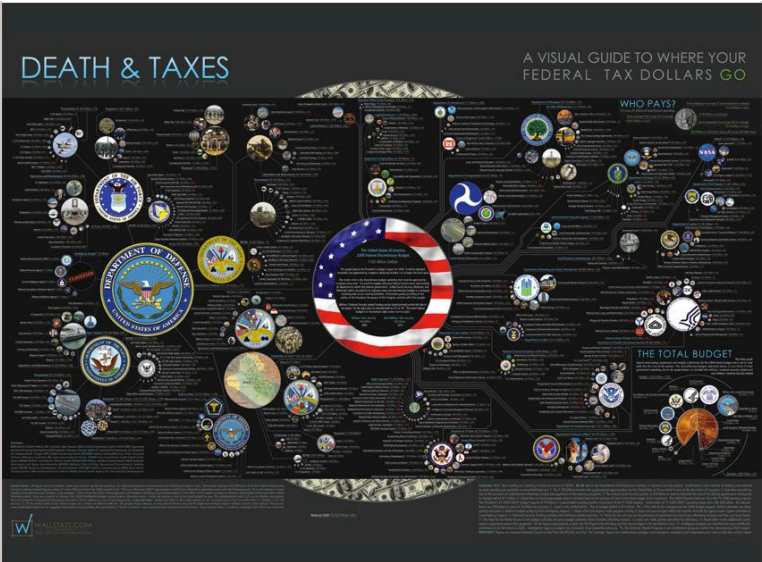


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Bachman, Jess. 2009. Death and Taxes. 2009. Burlington, VT. Courtesy of Jess Bachman. In Katy Börner & Elisha F. Hardy (Eds.), 5th Iteration (2009): Science Maps for Science Policy Makers, Places and Spaces: Mapping Science. <http://scimaps.org>

TAX DOLLARS AT WORK:

This conceptual map breaks down more than a trillion dollars in federal spending, as allocated by President George W. Bush's budget request for 2009. Military and national security spending dominate the budget, eating up more than two thirds of the allotted funds. (The map tracks only discretionary spending, which constitutes roughly one third of the total federal budget; the rest goes to mandatory spending on programs such as Social Security and Medicare.) The left side of the map primarily charts the distribution of monies among Department of Defense subdepartments and programs as well as to other projects relating to the war on terror. For a side-by-side comparison of Bush and his successor, [see the map author's 2011 version](#), based on President Barack Obama's most recent budget request.

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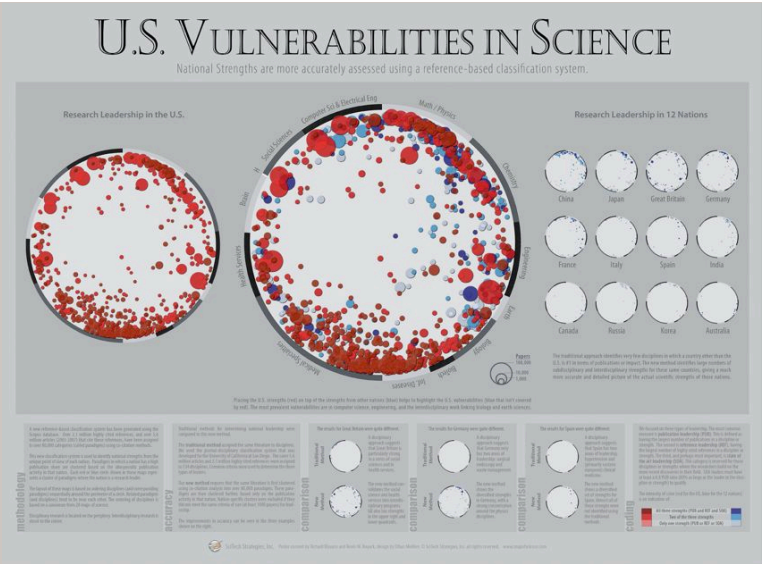
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Boyack, Kevin W. and Richard Klavans. 2008. U.S. Vulnerabilities in Science. Courtesy of SciTech Strategies. In Katy Börner & Elisha F. Hardy (Eds.), 5th Iteration (2009): Science Maps for Science Policy Makers, *Places and Spaces: Mapping Science*. <http://scimaps.org>

SOFT SPOTS:

The authors of this 2008 map assigned millions of scholarly papers from around the world to subject-matter groups, represented by colored dots, in order to identify the fields in which various countries have established leadership roles. The perimeter of the circular plots is divided into 13 broad subject areas; dots near the edge of the circle are largely confined to one field, whereas dots near the center are interdisciplinary. The smaller circular plot to the left shows U.S. research alone (red dots), whereas the main plot in the center merges U.S. research with that from 12 other nations (blue dots). The authors note that this visual assessment of the global science sphere highlights areas in which the U.S. is in danger of falling behind. "The most prevalent vulnerabilities," they wrote, "are in computer science, engineering, and the interdisciplinary work linking biology and Earth sciences."

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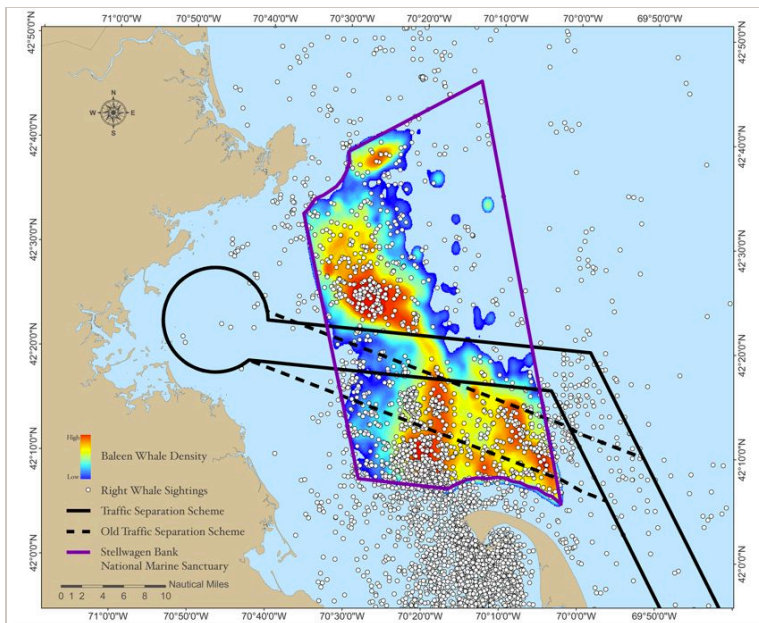
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Wiley, David N., Michael A. Thompson and Richard Merrick. 2006. Realigning the Boston Traffic Separation Scheme to Reduce the Risk of Ship Strike to Right and Other Baleen Whales. Boston. In Katy Börner & Elisha F. Hardy (Eds.), 5th Iteration (2009): Science Maps for Science Policy Makers, Places and Spaces: Mapping Science. <http://scimaps.org>

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WHALE-BEING:

This marine cartographic map depicts an area of potential collisions between oceangoing ships and whales, where the Boston Traffic Separation Scheme crosses the Stellwagen Bank National Marine Sanctuary. The map's authors plotted more than a quarter of a million whale sightings in the region, dating back to 1979, to estimate areas of higher and lower population density within the sanctuary. The waters are home to the endangered North Atlantic Right Whale, among other species. The researchers' data showed that by revising the shipping traffic scheme slightly (from the dotted lines to the solid lines), tacking on only nine to 22 minutes of transit time for ships entering or leaving Boston Harbor, whale sightings by ships passing through the sanctuary would be reduced by 81 percent. The authors report that their proposal was accepted by the United Nations International Maritime Organization in 2006 and took effect the following year.

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